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Systems and methods for producing aviation fuel

Abstract

Embodiments of systems and methods to produce aviation fuel are disclosed. An example of a method to produce aviation fuel includes fractionating a renewable diesel feedstock in a fractionator to produce a C.sub.8− fraction, a C.sub.8-18 fraction, and a C.sub.18+ fraction. Additionally, the method includes providing the C.sub.8-18 fraction to an isomerization reactor to produce an aviation fuel product. The method includes supplying at least a portion of the C.sub.18+ fraction to a hydrocracking reactor to produce a hydrocracked product. The method further includes recycling at least a portion of the hydrocracked product to the fractionator for fractionating along with the renewable diesel feedstock.

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6558531	12/2002	Steffens et al.	N/A	N/A
6589323	12/2002	Korin	N/A	N/A
6592448	12/2002	Williams	N/A	N/A
6609888	12/2002	Ingistov	N/A	N/A
6622490	12/2002	Ingistov	N/A	N/A
6644935	12/2002	Ingistov	N/A	N/A
6660895	12/2002	Brunet et al.	N/A	N/A
6672858	12/2003	Benson et al.	N/A	N/A
6733232	12/2003	Ingistov et al.	N/A	N/A
6733237	12/2003	Ingistov	N/A	N/A
6736961	12/2003	Plummer et al.	N/A	N/A
6740226	12/2003	Mehra et al.	N/A	N/A
6772581	12/2003	Ojiro et al.	N/A	N/A
6772741	12/2003	Pittel et al.	N/A	N/A
6814941	12/2003	Naunheimer et al.	N/A	N/A
6824673	12/2003	Ellis et al.	N/A	N/A
6827841	12/2003	Kiser et al.	N/A	N/A
6835223	12/2003	Walker et al.	N/A	N/A
6841133	12/2004	Niewiedzial et al.	N/A	N/A
6842702	12/2004	Haaland et al.	N/A	N/A
6854346	12/2004	Nimberger	N/A	N/A
6858128	12/2004	Hoehn et al.	N/A	N/A
6866771	12/2004	Lomas et al.	N/A	N/A
6869521	12/2004	Lomas	N/A	N/A
6897071	12/2004	Sonbul	N/A	N/A
6962484	12/2004	Brandl et al.	N/A	N/A
7013718	12/2005	Ingistov et al.	N/A	N/A
7035767	12/2005	Archer et al.	N/A	N/A
7048254	12/2005	Laurent et al.	N/A	N/A
7074321	12/2005	Kalnes	N/A	N/A
7078005	12/2005	Smith et al.	N/A	N/A
7087153	12/2005	Kalnes	N/A	N/A
7156123	12/2006	Welker et al.	N/A	N/A
7172686	12/2006	Ji et al.	N/A	N/A
7174715	12/2006	Armitage et al.	N/A	N/A
7194369	12/2006	Lundstedt et al.	N/A	N/A
7213413	12/2006	Battiste et al.	N/A	N/A
7225840	12/2006	Craig et al.	N/A	N/A
7228250	12/2006	Naiman et al.	N/A	N/A

7244350	12/2006	Kar et al.	N/A	N/A
7252755	12/2006	Kiser et al.	N/A	N/A
7255531	12/2006	Ingistov	N/A	N/A
7260499	12/2006	Watzke et al.	N/A	N/A
7291257	12/2006	Ackerson et al.	N/A	N/A
7332132	12/2007	Hedrick et al.	N/A	N/A
7404411	12/2007	Welch et al.	N/A	N/A
7419583	12/2007	Nieskens et al.	N/A	N/A
7445936	12/2007	O'Connor et al.	N/A	N/A
7459081	12/2007	Koenig	N/A	N/A
7485801	12/2008	Pulter et al.	N/A	N/A
7487955	12/2008	Buercklin	N/A	N/A
7501285	12/2008	Triche et al.	N/A	N/A
7551420	12/2008	Cerqueira et al.	N/A	N/A
7571765	12/2008	Themig	N/A	N/A
7637970	12/2008	Fox et al.	N/A	N/A
7669653	12/2009	Craster et al.	N/A	N/A
7682501	12/2009	Soni et al.	N/A	N/A
7686280	12/2009	Lowery	N/A	N/A
7857964	12/2009	Mashiko et al.	N/A	N/A
7866346	12/2010	Walters	N/A	N/A
7895011	12/2010	Youssefi et al.	N/A	N/A
7914601	12/2010	Farr et al.	N/A	N/A
7931803	12/2010	Buchanan	N/A	N/A
7932424	12/2010	Fujimoto et al.	N/A	N/A
7939335	12/2010	Triche et al.	N/A	N/A
7981361	12/2010	Bacik	N/A	N/A
7988753	12/2010	Fox et al.	N/A	N/A
7993514	12/2010	Schlueter	N/A	N/A
8007662	12/2010	Lomas et al.	N/A	N/A
8017910	12/2010	Sharpe	N/A	N/A
8029662	12/2010	Varma et al.	N/A	N/A
8037938	12/2010	Jardim De Azevedo et al.	N/A	N/A
8038774	12/2010	Peng	N/A	N/A
8064052	12/2010	Feitisch et al.	N/A	N/A
8066867	12/2010	Dziabala	N/A	N/A
8080426	12/2010	Moore et al.	N/A	N/A
8127845	12/2011	Assal	N/A	N/A
8193401	12/2011	McGehee et al.	N/A	N/A
8236566	12/2011	Carpenter et al.	N/A	N/A
8286673	12/2011	Recker et al.	N/A	N/A
8354065	12/2012	Sexton	N/A	N/A
8360118	12/2012	Fleischer et al.	N/A	N/A
8370082	12/2012	De Peinder et al.	N/A	N/A
8388830	12/2012	Sohn et al.	N/A	N/A
8389285	12/2012	Carpenter et al.	N/A	N/A
8397803	12/2012	Crabb et al.	N/A	N/A
8397820	12/2012	Fehr et al.	N/A	N/A
8404103	12/2012	Dziabala	N/A	N/A

8434800	12/2012	LeBlanc	N/A	N/A
8481942	12/2012	Mertens	N/A	N/A
8506656	12/2012	Turocy	N/A	N/A
8518131	12/2012	Mattingly et al.	N/A	N/A
8524180	12/2012	Canari et al.	N/A	N/A
8569068	12/2012	Carpenter et al.	N/A	N/A
8579139	12/2012	Sablak	N/A	N/A
8591814	12/2012	Hodges	N/A	N/A
8609048	12/2012	Beadle	N/A	N/A
8647415	12/2013	De Haan et al.	N/A	N/A
8670945	12/2013	van Schie	N/A	N/A
8685232	12/2013	Mandal et al.	N/A	N/A
8735820	12/2013	Mertens	N/A	N/A
8753502	12/2013	Sexton et al.	N/A	N/A
8764970	12/2013	Moore et al.	N/A	N/A
8778823	12/2013	Oyekan et al.	N/A	N/A
8781757	12/2013	Farquharson et al.	N/A	N/A
8784645	12/2013	Iguchi et al.	N/A	N/A
8829258	12/2013	Gong et al.	N/A	N/A
8916041	12/2013	Van Den Berg et al.	N/A	N/A
8932458	12/2014	Gianzon et al.	N/A	N/A
8986402	12/2014	Kelly	N/A	N/A
8987537	12/2014	Droubi et al.	N/A	N/A
8999011	12/2014	Stern et al.	N/A	N/A
8999012	12/2014	Kelly et al.	N/A	N/A
9011674	12/2014	Milam et al.	N/A	N/A
9057035	12/2014	Kraus et al.	N/A	N/A
9097423	12/2014	Kraus et al.	N/A	N/A
9109176	12/2014	Stern et al.	N/A	N/A
9109177	12/2014	Freel et al.	N/A	N/A
9138738	12/2014	Glover et al.	N/A	N/A
9216376	12/2014	Liu et al.	N/A	N/A
9272241	12/2015	Königsson	N/A	N/A
9273867	12/2015	Buzinski et al.	N/A	N/A
9279748	12/2015	Hughes et al.	N/A	N/A
9289715	12/2015	Høy-Petersen et al.	N/A	N/A
9315403	12/2015	Laur et al.	N/A	N/A
9371493	12/2015	Oyekan	N/A	N/A
9371494	12/2015	Oyekan et al.	N/A	N/A
9377340	12/2015	Hägg	N/A	N/A
9393520	12/2015	Gomez	N/A	N/A
9410102	12/2015	Eaton et al.	N/A	N/A
9428695	12/2015	Narayanaswamy et al.	N/A	N/A
9453169	12/2015	Stippich, Jr. et al.	N/A	N/A
9458396	12/2015	Weiss et al.	N/A	N/A
9487718	12/2015	Kraus et al.	N/A	N/A
9499758	12/2015	Droubi et al.	N/A	N/A
9500300	12/2015	Daigle	N/A	N/A
9506649	12/2015	Rennie et al.	N/A	N/A

9580662	12/2016	Moore	N/A	N/A
9624448	12/2016	Joo et al.	N/A	N/A
9650580	12/2016	Merdrignac et al.	N/A	N/A
9657241	12/2016	Craig et al.	N/A	N/A
9662597	12/2016	Formoso	N/A	N/A
9663729	12/2016	Baird et al.	N/A	N/A
9665693	12/2016	Saeger et al.	N/A	N/A
9709545	12/2016	Mertens	N/A	N/A
9757686	12/2016	Peng	N/A	N/A
9789290	12/2016	Forsell	N/A	N/A
9803152	12/2016	Kar et al.	N/A	N/A
9834731	12/2016	Weiss et al.	N/A	N/A
9840674	12/2016	Weiss et al.	N/A	N/A
9873080	12/2017	Richardson	N/A	N/A
9878300	12/2017	Norling	N/A	N/A
9890907	12/2017	Highfield et al.	N/A	N/A
9891198	12/2017	Sutan	N/A	N/A
9895649	12/2017	Brown et al.	N/A	N/A
9896630	12/2017	Weiss et al.	N/A	N/A
9914094	12/2017	Jenkins et al.	N/A	N/A
9920270	12/2017	Robinson et al.	N/A	N/A
9925486	12/2017	Botti	N/A	N/A
9982788	12/2017	Maron	N/A	N/A
9988585	12/2017	Hayasaka et al.	N/A	N/A
10018458	12/2017	Wade et al.	N/A	N/A
10047299	12/2017	Rubin-Pitel et al.	N/A	N/A
10048100	12/2017	Workman, Jr.	N/A	N/A
10087397	12/2017	Phillips et al.	N/A	N/A
10099175	12/2017	Takashashi et al.	N/A	N/A
10150078	12/2017	Komatsu et al.	N/A	N/A
10228708	12/2018	Lambert et al.	N/A	N/A
10239034	12/2018	Sexton	N/A	N/A
10253269	12/2018	Cantley et al.	N/A	N/A
10266779	12/2018	Weiss et al.	N/A	N/A
10295521	12/2018	Mertens	N/A	N/A
10308884	12/2018	Klussman	N/A	N/A
10316263	12/2018	Rubin-Pitel et al.	N/A	N/A
10384157	12/2018	Balcik	N/A	N/A
10435339	12/2018	Larsen et al.	N/A	N/A
10435636	12/2018	Johnson et al.	N/A	N/A
10443000	12/2018	Lomas	N/A	N/A
10443006	12/2018	Fruchey et al.	N/A	N/A
10457881	12/2018	Droubi et al.	N/A	N/A
10479943	12/2018	Liu et al.	N/A	N/A
10494579	12/2018	Wrigley et al.	N/A	N/A
10495570	12/2018	Owen et al.	N/A	N/A
10501699	12/2018	Robinson et al.	N/A	N/A
10526547	12/2019	Larsen et al.	N/A	N/A
10533141	12/2019	Moore et al.	N/A	N/A

10563130	12/2019	Narayanaswamy et al.	N/A	N/A
10563132	12/2019	Moore et al.	N/A	N/A
10563133	12/2019	Moore et al.	N/A	N/A
10570078	12/2019	Larsen et al.	N/A	N/A
10577551	12/2019	Kraus et al.	N/A	N/A
10584287	12/2019	Klussman et al.	N/A	N/A
10604709	12/2019	Moore et al.	N/A	N/A
10640719	12/2019	Freel et al.	N/A	N/A
10655074	12/2019	Moore et al.	N/A	N/A
10696906	12/2019	Cantley et al.	N/A	N/A
10808184	12/2019	Moore	N/A	N/A
10836966	12/2019	Moore et al.	N/A	N/A
10876053	12/2019	Klussman et al.	N/A	N/A
10954456	12/2020	Moore et al.	N/A	N/A
10961468	12/2020	Moore et al.	N/A	N/A
10962259	12/2020	Shah et al.	N/A	N/A
10968403	12/2020	Moore	N/A	N/A
11021662	12/2020	Moore et al.	N/A	N/A
11098255	12/2020	Larsen et al.	N/A	N/A
11124714	12/2020	Eller et al.	N/A	N/A
11136513	12/2020	Moore et al.	N/A	N/A
11164406	12/2020	Meroux et al.	N/A	N/A
11168270	12/2020	Moore	N/A	N/A
11175039	12/2020	Lochschmied et al.	N/A	N/A
11200489	12/2020	Cohen et al.	N/A	N/A
11203719	12/2020	Cantley et al.	N/A	N/A
11203722	12/2020	Moore et al.	N/A	N/A
11214741	12/2021	Davdov et al.	N/A	N/A
11306253	12/2021	Timken et al.	N/A	N/A
11319262	12/2021	Wu et al.	N/A	N/A
11352577	12/2021	Woodchick et al.	N/A	N/A
11352578	12/2021	Eller et al.	N/A	N/A
11384301	12/2021	Eller et al.	N/A	N/A
11421162	12/2021	Pradeep et al.	N/A	N/A
11460478	12/2021	Sugiyama et al.	N/A	N/A
11467172	12/2021	Mitzel et al.	N/A	N/A
11494651	12/2021	Mukund et al.	N/A	N/A
11542441	12/2022	Larsen et al.	N/A	N/A
11574192	12/2022	Cohen et al.	N/A	N/A
11578638	12/2022	Thobe	N/A	N/A
11634647	12/2022	Cantley et al.	N/A	N/A
11667858	12/2022	Eller et al.	N/A	N/A
11692141	12/2022	Larsen et al.	N/A	N/A
11702600	12/2022	Sexton et al.	N/A	N/A
11715950	12/2022	Miller et al.	N/A	N/A
11720526	12/2022	Miller et al.	N/A	N/A
11802257	12/2022	Short et al.	N/A	N/A
11835450	12/2022	Bledsoe, Jr. et al.	N/A	N/A
11860069	12/2023	Bledsoe, Jr.	N/A	N/A

11886154	12/2023	Mukund et al.	N/A	N/A
11891581	12/2023	Cantley et al.	N/A	N/A
11898109	12/2023	Sexton et al.	N/A	N/A
11905468	12/2023	Sexton et al.	N/A	N/A
11905479	12/2023	Eller et al.	N/A	N/A
11906423	12/2023	Bledsoe, Jr. et al.	N/A	N/A
11920096	12/2023	Woodchick et al.	N/A	N/A
11921035	12/2023	Bledsoe, Jr. et al.	N/A	N/A
11970664	12/2023	Larsen	N/A	N/A
11975316	12/2023	Zalewski	N/A	N/A
11993751	12/2023	Jagnanan et al.	N/A	N/A
12000720	12/2023	Langlois, III	N/A	N/A
12018216	12/2023	Larsen et al.	N/A	N/A
12031094	12/2023	Sexton et al.	N/A	N/A
12031676	12/2023	Craig et al.	N/A	N/A
12037548	12/2023	Larsen et al.	N/A	N/A
12039446	12/2023	Cohen et al.	N/A	N/A
12049592	12/2023	Clark et al.	N/A	N/A
12066800	12/2023	Salhov et al.	N/A	N/A
12163878	12/2023	Bledsoe, Jr.	N/A	N/A
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to, and the benefit of U.S. Provisional Application No. 63/548,081, filed Nov. 10, 2023, titled “SYSTEMS AND METHODS FOR PRODUCING AVIATION FUEL,” the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD
(1) The disclosure is generally related to aviation fuel. More specifically, the disclosure relates to systems and methods for producing aviation fuels, such as synthetic paraffinic kerosene (SPK) or sustainable aviation fuel (SAF).
BACKGROUND
(2) An increasing number of facilities are implementing renewable feedstock or materials to provide sustainable transportation fuels, which may include a reduced environmental impact compared to fossil-based fuels. Certain aviation fuels may be developed from renewable feedstocks to include chemical properties suitable for powering aircrafts. In some cases, these fuels are referred to as synthetic paraffinic kerosene (SPK), which can be used directly as a sustainable aviation fuel (SAF) or blended with any other fuel or product to produce SAF having desired properties. Traditional process flow schemes for creating aviation fuel include directing a feedstock into an isomerization unit, then a fractionation unit, and then a hydrocracking unit. This process flow therefore directs all compounds of the feedstock directly into the isomerization unit, including large-size compounds that are subsequently cracked in the hydrocracking unit for size reduction. Isomerization of these compounds negatively contribute to the operating conditions and design considerations of various components, such as the hydrocracking unit. For example, certain situations necessitate management of molecular sizes and freeze points of the various compounds for both the isomerization unit and the hydrocracking unit, which are overly complex and problematic. Moreover, single pass hydrocracking in the traditional process flow schemes require removal of a drag stream of unconverted compounds that are too large or otherwise unsuitable for

jet fuel use. The drag stream, which may be supplied or retailed as renewable diesel, further reduces yield or efficiency of aviation fuel production.

SUMMARY

(3) Accordingly, Applicant has recognized a need for systems and methods to provide enhanced yield of aviation fuel from a renewable diesel feedstock. The present disclosure is directed to embodiments of systems and methods for producing aviation fuel.

(4) Embodiments of refining systems disclosed herein include a configuration of process units for an enhanced yield of aviation fuel, such as synthetic paraffinic kerosene (SPK) or sustainable aviation fuel (SAF). In some examples, SPK is a blendstock that is blended with an aromatic-rich fuel according to one or more ASTM standards to produce SAF having specified properties. For example, the SPK may be blended with a sustainably sourced fuel and/or a petroleum-derived fuel to produce SAF that includes a threshold content of aromatic compounds therein. As used herein, the aviation fuel produced according to the present techniques refers to any suitable SPK, SAF, or renewable jet fuel that includes physical and chemical properties suitable for aviation use. The presently disclosed refining systems provide a disruption and reversal of the conventional order of dewaxing reactions (hydrocracking and isomerization) and the fractionator to provide a more efficient processing of the feedstock. The systems facilitate simpler and potentially lower severity in operation of the hydrocracking and isomerization units for increased yield. For example, compounds that are outside the boiling point ranges of aviation fuel are recycled to extinction, thereby providing aviation fuel as one of the main distillate products. In certain examples, the aviation fuel is the main distillate product. In an example, the system includes the fractionator and hydrocracking reactor being positioned upstream of the isomerization reactor to drive product separation; thereby providing only specific, targeted compounds to each reaction step. In certain examples, this configuration reduces over-cracking that produces undesired light ends, maximizing the aviation fuel yield. The configuration also enables each reaction step (hydrocracking and isomerization) to have a focused processing operation. The refining systems and methods disclosed herein separate normal alkanes into fractions that are supplied to their respective upgrading processes to modify molecular sizes and qualities for increased production of aviation fuel.

(5) An embodiment of the disclosure is directed to a method to produce aviation fuel. The method may include fractionating a renewable diesel feedstock in a fractionator to produce a C.sub.8− fraction, a C.sub.8-18 fraction, and a C.sub.18, fraction. The C.sub.8− fraction is further processed to produce naphtha, LPG, or fuel gas (collectively referred to as light ends). Additionally, the method includes providing the C.sub.8-18 fraction to an isomerization reactor to produce an aviation fuel product. The method also includes supplying at least a portion of the C.sub.18+ fraction to a hydrocracking reactor to produce a hydrocracked product and recycling at least a portion of the hydrocracked product to the fractionator for fractionating along with the renewable diesel feedstock.

(6) In another example, the renewable diesel feedstock includes a hydrodeoxygenated renewable diesel. In another example, the renewable diesel feedstock contains normal alkanes produced by hydrodeoxygenation of glycerides. In another example, the method may further include hydrotreating a renewable feedstock containing triglycerides, triglyceride derivatives, or a combination thereof in a hydrogen-rich atmosphere to produce the renewable diesel feedstock.

(7) In another example, the method may further include mixing the hydrocracked product with the renewable diesel feedstock directly upstream of the fractionator. In another example, the renewable diesel feedstock is supplied to a first inlet of the fractionator and the at least a portion of the hydrocracked product is supplied to a second inlet of the fractionator. In another example, the recycling at least a portion of the hydrocracked product to the fractionator further includes recycling an entirety of the hydrocracked product. In another example, the method may further include producing an additional C.sub.8− fraction in the isomerization reactor, providing the aviation fuel product and the additional C.sub.8− fraction to a separator to separate the aviation fuel

product from the additional C.sub.8− fraction. The method may include recycling the additional C.sub.8− fraction to the fractionator. The method may include further processing of the C.sub.8− fraction to produce a plurality of light ends. In another example, the method may further include selecting an operating temperature of the hydrocracking reactor. The operating temperature may be lower than an operating temperature of a hydrocracking reactor positioned downstream of a reference isomerization reactor.
(8) Another embodiments of the disclosure is directed to a method to produce aviation fuel. The method may include supplying a hydrodeoxygenated renewable diesel feedstock to a fractionator. The method includes fractionating the hydrodeoxygenated renewable diesel feedstock into a C.sub.8− fraction, a C.sub.8-18 fraction, and a C.sub.18+ fraction. The method includes providing the C.sub.8-18 fraction to an isomerization reactor to produce an aviation fuel product. The method includes supplying at least a portion of the C.sub.18+ fraction to a hydrocracking reactor to produce a hydrocracked product. The hydrocracking reactor is configured to crack one or more heavy compounds of the C.sub.18+ fraction into two or more lighter compounds. The method also includes recycling at least a portion of the hydrocracked product to the fractionator.
(9) In another example, the method may further include hydrodeoxygenating a triglyceride feedstock to produce the hydrodeoxygenated renewable diesel feedstock. In another example, the method may further include producing an additional C.sub.8− fraction in the isomerization reactor, providing the aviation fuel product and the additional C.sub.8− fraction to a separator to separate the aviation fuel product from the additional C.sub.8− fraction, and recycling the additional C.sub.8− fraction to the fractionator.
(10) Another embodiment of the disclosure is directed to a system to produce an aviation fuel. The system includes a fractionator having a first inlet to receive a renewable diesel feedstock, a first outlet, a second outlet, and a third outlet. The fractionator is configured to fractionate the renewable diesel feedstock into a C.sub.8− fraction output that exits through the first outlet, a C.sub.8-18 fraction output to the second outlet, and a C.sub.18+ fraction output to the third outlet. The system also includes a hydrocracking reactor having a second inlet and a fourth outlet. The second inlet is connected to and in fluid communication with the third outlet to receive the C.sub.18+ fraction. The hydrocracking reactor is configured to hydrocrack the C.sub.18+ fraction into a hydrocracked product. The fourth outlet is in fluid communication with the fractionator and facilitates recycling of the hydrocracked product to the fractionator. Additionally, the system may include an isomerization reactor having a third inlet and a fifth outlet. The third inlet may be connected to and in fluid communication with the second outlet to receive the C.sub.8-18 fraction. The isomerization reactor may be operable to isomerize the C.sub.8-18 fraction and supply an aviation fuel product as output through the fifth outlet.
(11) In certain examples, the renewable diesel feedstock contains normal alkanes produced by hydrodeoxygenation of glycerides, free fatty acids, or combinations thereof. In some examples, a majority of the hydrodeoxygenated renewable diesel contains C.sub.16 and C.sub.18 normal alkanes. In some examples, the system may include a separator having a fifth inlet connected to and in fluid communication with the fifth outlet, a sixth outlet to supply an additional C.sub.8− fraction to the fractionator, and a seventh outlet to output the aviation fuel product. In another example, the fourth outlet is connected to and in fluid communication with the first inlet or an additional inlet of the fractionator. In another example, the system may include a hydrogen source connected to and in fluid communication with the hydrocracking reactor, the isomerization reactor, or both.
(12) In some examples, the system may include a controller in signal communication with the fractionator, the hydrocracking reactor, and the isomerization reactor. The controller may be configured to transmit signals to one or more of the fractionator, the hydrocracking reactor, and the isomerization reactor to adjust production of the aviation fuel product. In some examples, the controller is configured to adjust operation of the fractionator, the hydrocracking reactor, or both in response to detecting that the isomerization reactor is deviating from one or more preselected

operating parameter ranges.
(13) Another embodiment of the disclosure is directed to a controller to produce aviation fuel. The controller includes a memory storing processor-executable instructions. The controller also includes one or more processors communicatively coupled to the memory and configured to execute the processor-executable instructions from the memory. The one or more processors are configured, when executing the instructions, to transmit first signals to one or more first actuators associated with a fractionator to control operation thereof. The first signals cause the fractionator to fractionate a renewable diesel feedstock to produce a C.sub.8− fraction, a C.sub.8-18 fraction, and a C.sub.18+ fraction. The one or more processors are configured, when executing the instructions, to transmit second signals to one or more second actuators associated with an isomerization reactor to control operation thereof. The second signals cause the isomerization reactor to receive the C.sub.8-18 fraction and produce an aviation fuel product. Additionally, the one or more processors are configured, when executing the instructions, to transmit third signals to one or more third actuators associated with a hydrocracking reactor to control operation thereof. The third signals cause the hydrocracking reactor to receive at least a portion of the C.sub.18+ fraction, produce a hydrocracked product, and direct at least a portion of the hydrocracked product in a recycle stream to the fractionator for fractionating along with the renewable diesel feedstock.
(14) In certain examples, the one or more processors are configured, when executing the instructions, to control operation of the fractionator to within one or more first preselected operating parameter ranges, control operation of the isomerization reactor to within one or more second preselected operating parameter ranges, and control operation of the hydrocracking reactor to within one or more third preselected operating parameter ranges.
(15) In some examples, the one or more processors are configured, when executing the instructions, to adjust operation of the hydrocracking reactor in response to detecting that the fractionator is deviating from the first preselected operating parameter ranges, and adjust operation of the fractionator in response to detecting that the hydrocracking reactor is deviating from the third preselected operating parameter ranges. In an example, the one or more processors are configured, when executing the instructions, to adjust operation of the fractionator, the hydrocracking reactor, or both in response to detecting that the isomerization reactor is deviating from the second preselected operating parameter ranges.
(16) In certain examples, the one or more processors are configured, when executing the instructions, to update one or more of the first preselected operating parameter ranges, the second preselected operating parameter ranges, and the third preselected operating parameter ranges in response to receiving a respective, modified operating parameter range from an authorized computing device.
(17) In another example, the controller is in signal communication with one or more sensors configured to provide sensor data associated with one or more operating parameters of the fractionator, the isomerization reactor, and the hydrocracking reactor. In some examples, one or more of the first actuators, the second actuators, and the third actuators include a control valve, a pump, a compressor, a heating element, or an electrical switch.
(18) Still other aspects and advantages of these and other examples are discussed in detail herein. Moreover, it is to be understood that both the foregoing information and the following detailed description provide merely illustrative examples of various aspects and examples and are intended to provide an overview or framework for understanding the nature and character of the claimed aspects and examples. Accordingly, the advantages and features of the present disclosure will become more apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood that the features of the various examples described herein are not mutually exclusive and may exist in various combinations and permutations.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) These and other features, aspects, and advantages of the disclosure will become better understood with regard to the following descriptions, claims, and accompanying drawings. It is to be noted, however, that the drawings illustrate only several examples of the disclosure and, therefore, are not to be considered limiting of the scope of the disclosure.

(2) FIG. **1** is a schematic diagram of a refining system for producing aviation fuel according to embodiments of the disclosure;

(3) FIG. **2** is an illustration of a method for producing aviation fuel, such as with the refining system of FIG. **1**, according to embodiments of the disclosure;

(4) FIG. **3** is a schematic diagram of another refining system for producing aviation fuel according to embodiments of the disclosure;

(5) FIG. **4** is an illustration of a method for producing aviation fuel, such as with the refining system of FIG. **3**, according to embodiments of the disclosure;

(6) FIG. **5** is a schematic diagram of a controller of a refining system according to embodiments of the disclosure; and

(7) FIG. **6** is an illustration of a method for producing aviation fuel as performed by a controller according to embodiments of the disclosure.

DETAILED DESCRIPTION

(8) So that the manner in which the features and advantages of the examples of the systems and methods disclosed herein, as well as others, which will become apparent, may be understood in more detail, a more particular description of examples of systems and methods briefly summarized above may be had by reference to the following detailed description of examples thereof, in which one or more are further illustrated in the appended drawings, which form a part of this specification. It is to be noted, however, that the drawings illustrate only various examples of the examples of the systems and methods disclosed herein and are therefore not to be considered limiting of the scope of the systems and methods disclosed herein as it may include other effective examples as well.

(9) As used herein, the term “C.sub.x-y compounds,” in which x and y are positive integer values, refers to hydrocarbon-based compounds, each compound containing between x and y carbon atoms, x and y inclusive. For example, a C.sub.3-5 fraction or stream refers to a mixture that substantially contains or entirely contains hydrocarbon-based compounds, each compound containing 3, 4, or 5 carbon atoms. Additionally, it may be noted that, in certain cases, a C.sub.x-y fraction may not include a respective compound having each of the referenced integer values. As one example, a C.sub.4-8 fraction can be a stream that contains compounds of 4, 5, and 7 carbon atoms, without any compounds of 6 or 8 carbon atoms. As used herein, the term “C.sub.x+ compounds,” in which x is a positive integer value, refers to hydrocarbon-based compounds, each compound containing at least x carbon atoms. For example, a C.sub.3+ fraction refers to a mixture that substantially contains or entirely contains hydrocarbon-based compounds, each compound containing 3 or more (e.g., 3, 4, 5, 6, and so forth) carbon atoms.

(10) As used herein, the term “C.sub.x− compounds,” in which x is a positive integer value, refers to hydrocarbon-based compounds, each compound containing no more than x carbon atoms. For example, a C.sub.4− fraction refers to a mixture that substantially contains or entirely contains hydrocarbon-based compounds, each compound containing 4, 3, 2, or 1 carbon atoms. It may be noted that, in certain cases, a “C.sub.x− fraction” may also include hydrogen (H.sub.2), in addition to hydrocarbons having x or fewer carbon atoms. The term “substantially contains” means that the mixture includes more than 50%, or at least 51%, or at least 60%, or at least 70%, or at least 80% by weight of the relevant hydrocarbon-based compounds.

(11) As used herein, the terms “hydrocracking reactor” and “isomerization reactor” can each refer to an area including one or more units, zones, or sub-zones, which can each perform a respective unit operation and collectively operate as a reactor. The reactors can include one or more reactors or reactor vessels, separators, strippers, flash drums, extraction columns, fractionation columns, heaters, coolers, exchangers, pipes, pumps, compressors, and controllers. As an example, a hydrocracking reactor or an isomerization reactor can include one or more heaters, one or more coolers, one or more separators, and/or a stripper. Additionally, a unit, such as a reactor, dryer, or vessel, can further include one or more zones or sub-zones that contain various equipment. In examples, a flash drum or flash includes relatively simple equipment to facilitate efficient separations between hydrocarbon-based compounds supplied thereto. In examples, a stripper includes more complex equipment than a flash drum and receives a stripping gas that facilitates more precise separations between hydrocarbon-based compounds supplied thereto. As used herein, the term “aviation fuel” refers to any suitable synthetic paraffinic kerosene (SPK), sustainable aviation fuel (SAF), or renewable jet fuel.
(12) As transportation industries move to reduce their carbon footprints, sustainable transportation fuels are becoming even more significant. To this end, certain facilities may implement renewable feedstocks as a replacement for all or a portion of petroleum products that are included in transportation fuels. For example, SAF or an SPK constituent thereof may be produced from a growing list of renewable or sustainable resources, such as fats, oils, greases, grains, seeds, waste streams, and so forth. However, in the emerging sustainability-focused environment, a low or insufficient yield of useable fuels may be produced from current refining practices of sustainable resources, thus increasing the amount of the resources that are utilized. As such, Applicant has recognized a need for systems and methods to provide aviation fuel at enhanced yields and efficiency, such as by improving or optimizing a configuration of unit operations for sustainable refining systems.
(13) In some embodiments, renewable diesel for aviation fuel production may be produced by hydrotreating renewable feedstocks containing triglycerides such as fats, oils, or greases, and free fatty acids derived from those triglycerides. Certain hydrodeoxygenation (HDO) reactions produce renewable diesel that may primarily contain C.sub.16 and C.sub.18 normal alkanes. Competing reactions may also produce byproducts including C.sub.15 and C.sub.17 normal alkanes. Minor byproducts of C.sub.14− and C.sub.19+ normal alkanes may also be formed during HDO reactions, depending upon the fatty acid distributions of the renewable feedstocks used. The resulting HDO renewable diesel feedstock or mixture may be processed into aviation fuel by (1) fractionating compounds having a proper boiling point for aviation fuel use (nominally C.sub.8-18 compounds) from smaller C.sub.8− compounds and larger C.sub.18+ compounds in the mixture, and (2) isomerizing aviation fuel-range compounds to lower a freeze point of the mixture to suitable levels within existing and/or renewable aviation fuel specifications.
(14) Embodiments of refining systems disclosed here position a fractionator as a front unit in the system, rather than at the end, for better flow management and efficient separation of the desired aviation fuel product from compounds that are too small or too large to be used as aviation fuel. Material that includes a higher boiling point than aviation fuel, such as a C.sub.18+ fraction, is obtained as the bottom product from the fractionator. This configuration of the refining system thus directs only the C.sub.18+ fraction to the hydrocracking reactor. As presently recognized, the yield of aviation fuel can be maximized by hydrocracking those C.sub.18+ compounds into a plurality of aviation fuel range compounds and smaller compounds corresponding to a naphtha, liquefied petroleum gas (LPG), and/or fuel gas size range, also denoted as light ends. In other words, the C.sub.18+ fraction may be fed or supplied to a hydrocracking reactor to crack it into a C.sub.8-18 product in the aviation fuel range and a lighter C.sub.8− compound in the light end range.
(15) In some embodiments, the hydrocracked products from the hydrocracking step are recycled to the fractionator as feed. Thus, the C.sub.18+ compounds that are too large for use as aviation fuel

can be recycled to extinction, thereby converting substantially all feedstock into an aviation fuel or lighter products that may have higher value. Additionally, the normal alkanes of the C.sub.18+ fraction, having not been processed within the isomerization reactor, may diffuse into and out of the pores of a zeolite hydrocracking catalyst easier than the corresponding branched compounds. This efficiency enables lower cracking temperatures to perform compound size reduction, compared to prior art systems without the present unit configuration. In other words, the refining system may increase or maximize aviation fuel production by converting substantially all of the C.sub.18+ fraction material into lower boiling point compounds that are suitable for use in aviation fuel. The compound size adjustment may therefore include clear benefits over other systems that may not fully utilize the heavier compounds of the feedstock for aviation fuel production.
(16) The C.sub.8-18 fraction from the fractionator may be sent to an isomerization reactor to lower its freeze point to meet aviation fuel specifications by converting the normal alkanes into branched or iso-alkanes of the same molecular weight. Additionally, any light ends produced by cracking in the isomerization reactor may be separated from the aviation fuel product and returned to the fractionator. As such, light ends from the fractionator are not processed in the isomerization reactor. Therefore, the isomerization reactor only receives the C.sub.8-18 fraction for isomerization. This allows for lower severity operation of the isomerization reactor and implementation of a reduced catalyst volume, compared to systems without the present unit configuration. Lower temperatures for both hydrocracking and isomerization steps limit or reduce production of light end byproducts, thus maximizing the yield of the most desirable aviation fuel product. As such, the present disclosure may separate normal alkanes into fractions suitable for their respective upgrading processes, thus efficiently modifying their molecular sizes and qualities, including boiling points, for maximized production of aviation fuel.
(17) FIG. **1** is a schematic diagram of a refining system **100** for producing aviation fuel, according to some embodiments disclosed herein. The refining system **100**, for example, may include multiple components or units that are optimally arranged to facilitate production of aviation fuel at higher efficiencies and yields than previously available systems. For example, the illustrated example of the refining system **100** includes a fractionator **120**, a hydrocracking reactor **130**, and an isomerization reactor **140**. As will be understood, the present configuration of and interconnections in the refining system **100** provide improved management of compounds in which compounds over a threshold size are converted to sizes suitable for aviation fuel use and only compounds sized for aviation fuel are supplied to the isomerization reactor **140**. As one example, positioning the fractionator **120** upstream of the hydrocracking reactor **130** and the isomerization reactor **140** facilitates these and other benefits disclosed herein.
(18) The refining system **100** utilizes a renewable diesel feedstock **118** as a feedstock for producing aviation fuel. The renewable diesel feedstock **118** may be a hydrodeoxygenated (HDO) renewable diesel feedstock, including normal alkanes produced by hydrodeoxygenation of glycerides, such as diglycerides, triglycerides, and/or their derivatives. For example, the renewable diesel feedstock **118** is produced by hydrotreating triglycerides and/or triglyceride derivatives, such as fats, oils, greases, and/or free fatty acids. This oxygen removal mechanism generally removes undesired oxygen from feed materials as H.sub.2O, thus conserving the carbon content of the renewable diesel feedstock **118**. In certain examples, the production of the renewable diesel feedstock **118** includes additional oxygen removal mechanisms, such as decarboxylation that removes CO.sub.2 and/or decarbonylation that removes CO. In some examples, the hydrodeoxygenation of triglycerides also causes decarboxylation and/or decarbonylation to occur. As such, references herein to hydrodeoxygenation also include any suitable decarboxylation and/or decarbonylation that occurs along with the hydrodeoxygenation. The renewable diesel feedstock **118** of certain examples is hydrotreated by hydrodeoxygenation in a respective, hydrogen-rich atmosphere to produce the renewable diesel feedstock.
(19) The examples of hydrodeoxygenation of triglycerides implemented herein generate alkanes

that, unlike free fatty acids, do not include a carboxyl functional group that entail further processing. As such, the renewable diesel feedstock **118** contains normal alkanes or straight-chain hydrocarbons ranging from C.sub.1 to C.sub.18+, In certain examples, a majority of the renewable diesel feedstock contains C.sub.16 and C.sub.18 normal alkanes. Use of alkanes in the fractionator **120** also provides further versatility to processing steps, such as by enabling later discussed recycle loops to seamlessly integrate with the feed to the fractionator. Additionally, the lower boiling point temperatures of alkanes compared to fatty free acids enables fractionation to proceed at lower temperatures, such that the fractionator **120** operates with lower temperatures and reduced energy usage, as compared to a comparable fractionator operating with free fatty acids.
(20) The renewable diesel feedstock **118** is supplied to the fractionator **120**, which separates the renewable diesel feedstock **118** into fractions or streams of alkanes having specific properties, such as a range of distillation temperatures or boiling points. The fractionator **120** includes a first vessel that has a first inlet, a first outlet, a second outlet, and a third outlet. In certain examples, the fractionator **120** operates in a continuous process that separates or divides the hydrocarbons therein based on pre-determined cut points established based on physical properties, such as boiling points or sizes. In particular, the fractionator **120** and/or renewable diesel feedstock **118** receive thermal energy from a heat source to generate a thermal gradient within the first vessel, which ranges from hottest temperatures near the bottom to coldest near the top of the first vessel. In some examples, the heat source includes a preheater, a heater, a fired heater, a furnace, and/or a reboiler. The hydrocarbons within the first vessel therefore vaporize at respective boiling points and rise until met with a region having a respective condensation temperature, at which a fraction having a desired composition can be removed.
(21) In the illustrated embodiment, the fractionator **120** separates the renewable diesel feedstock **118** into three fractions: a light fraction **122** having substantially C.sub.1-8 or C.sub.8− compounds, a middle fraction or aviation fuel-range fraction **124** having substantially C.sub.8-18 compounds, and a heavy fraction **126** having substantially C.sub.18+ compounds. These components are also referred to as a C.sub.8− fraction, a C.sub.8-18 fraction, and a C.sub.18+ fraction, in examples. The aviation fuel-range fraction **124** contains compounds with a suitable boiling point temperature for use as aviation fuel, such as sustainable aviation fuel and/or renewable jet fuel. In examples, the fractionator **120** separates the compounds within a threshold range of boiling point temperatures, associated with certain C.sub.8-18 compounds, and outputs them through the second outlet for further processing into aviation fuel. The light fraction **122** contains compounds with boiling point temperatures that are below the threshold range and thus can be provided as naphtha, LPG, and/or fuel gas by the refining system **100**. The light fraction **122** is continuously removed or sent through the first outlet of the fractionator **120**, in certain examples. The heavy fraction **126**, containing compounds with boiling point temperatures that are above the threshold range suitable for use as aviation fuel, can be continuously removed through the third outlet of the fractionator **120**.
(22) As recognized herein, the present refining system **100** upgrades, or adds value to, compounds of the heavy fraction **126** within the hydrocracking reactor **130** to increase a yield of aviation fuel. The hydrocracking reactor **130** includes a second vessel that has a second inlet and a fourth outlet. In examples, the second inlet of the hydrocracking reactor **130** is connected to and in fluid communication with the third outlet of the fractionator **120** to receive at least a portion of the heavy fraction **126** therefrom. In some examples, the hydrocracking reactor **130** receives an entirety of the heavy fraction **126**. In certain examples, a drag stream is removed from the heavy fraction **126** upstream of the hydrocracking reactor **130**, which receives a remaining portion of the heavy fraction **126**. Certain drag streams removed from the heavy fraction **160** are sold or utilized as a diesel product, in some examples. Within the hydrocracking reactor **130**, the heavy fraction **126** containing C.sub.18+ compounds are subject to hydrocracking to produce a hydrocracked product **132** that contains any suitable alkanes, such as C.sub.1-18+ compounds. Certain C.sub.18+ compounds of the heavy fraction **126** are cracked within the hydrocracking reactor **130** into a first

portion of compounds in the C.sub.8-18 range and a second portion of compounds in the C.sub.8− range.
(23) In more detail, compounds of the heavy fraction **126** are converted into two or more respective, smaller compounds when exposed to operating conditions in the hydrocracking reactor **130**. The operating conditions of the hydrocracking reactor **130** may include a high temperature and a high hydrogen pressure, provided with a zeolite hydrocracking catalyst. The hydrocracking catalyst of some examples includes a precious metal on a zeolite catalyst. In some examples, the hydrocracking catalyst is not sulfided, thus facilitating direct supply of the hydrocracked product **132** to the fractionator **120** (after any cooling and/or separations from recycle gas), without requiring use of hydrogen sulfide gas. In some examples, avoiding the use of hydrogen sulfide gas is desirable for systems including an isomerization catalyst containing precious metal because the sulfur would otherwise poison or negatively affect the precious metal of the isomerization catalyst. In some examples, the isomerization catalyst can include a base metal instead of a precious metal and thus be unaffected by hydrogen sulfide gas, such that the hydrocracking catalyst can be sulfided without negative effects.
(24) Additionally, to facilitate the hydrocracking reactions, the hydrocracking reactor **130** can have an operating temperature in a range between about 400 and about 800° F. (between about 204.4 and about 426.7° C.). In some examples, the operating temperature is in a range between about 300 and about 700° F. (between about 148.9 and about 371.1° C.). In some examples, the operating temperature is in a range between about 500 and about 900° F. (between about 260.0 and about 482.2° C.). The partial pressure of hydrogen in the hydrocracking reactor **130** is at least 200 psi, in some examples. In some examples, the partial pressure of hydrogen is at least 300, 400, 500, 600, 700, 800, 900, 1,000, 1,200, 1,500, 2,000, 2,500, or more psi. The hydrogen supplied to the hydrocracking reactor **130** may be provided by any suitable hydrogen source. It should be understood that, in some examples, one or more C.sub.18+ compounds may remain in the hydrocracked product **132**. For example, certain compounds produced from cracking large C.sub.19+ compounds include C.sub.18+ compounds and/or not all C.sub.18+ compounds of the heavy fraction **126** may be cracked in a first pass through the hydrocracking reactor **130**. As such, the hydrocracked product **132** of certain examples includes a wide variety of alkanes (e.g., C.sub.1-18+ compounds), while the heavy fraction **126** contains substantially C.sub.18+ compounds and excludes substantially C.sub.1-17 compounds.
(25) Compared to certain traditional operations, the disclosed refining system **100** having the fractionator **120** upstream of both the hydrocracking reactor **130** and the isomerization reactor **140** enables operating conditions of the hydrocracking reactor **130** to be customized and/or optimized for cracking the heavy fraction **126**. For example, the heavy fraction **126** supplied to the hydrocracking reactor **130** may include only C.sub.18+ normal alkanes. The normal alkanes diffuse into and out of pores of the zeolite hydrocracking catalyst more easily or efficiently than branched compounds, enabling lower cracking temperatures to be used for compound size reduction in the hydrocracking reactor **130**. In examples, the lower temperature limits or reduces production of light end byproducts, thus increasing or maximizing the yield of the more desirable aviation fuel product. The aviation fuel yield is further improved because the refining system **100** does not direct compounds already within the desired C.sub.8-18 range to hydrocracking. In other words, over-cracking that otherwise produces undesired light ends is reduced or avoided. The refining system **100** also reduces or minimizes a reactor size and corresponding capital expenses implemented for hydrocracking, because only the specific C.sub.18+ compounds to be reduced are supplied to the hydrocracking reactor **130**. Compared to higher temperature reactions, the lower temperatures utilized by the refining system **100** decrease the rate of undesired thermal cracking reactions that would otherwise lower yield of desired products and increase the rate of coke formation on the catalyst. Reduced coke formation is desired based on coke production reducing desired product yield via use of carbon content for forming coke, not aviation fuel, and deactivating the catalyst

more rapidly, thus requiring more frequent catalyst changes and consequently more process down time.
(26) The fourth outlet of the hydrocracking reactor **130** is connected to recycle at least a portion of the hydrocracked product **132** to the fractionator **120** or its feed. For example, the fourth outlet is connected to and in fluid communication with the first inlet of the fractionator **120**. The hydrocracked product **132** is therefore mixed with the renewable diesel feedstock **118** that is fed to the fractionator **120**. In other examples, the fourth outlet may be connected to another inlet of the fractionator **120**, as discussed with reference to FIG. **3** below. The fractionator **120** receives an entirety of the hydrocracked product **132** or a portion thereof, in certain examples. The hydrocracked product **132** of some examples is cooled and separated from recycle gas of the hydrocracking reactor **130** before entering the fractionator **120**. In any case, the refining system **100** includes recycling the hydrocracked product **132** to the fractionator **120**. As such, all compounds that have boiling points that are above boiling points for aviation fuel are recycled to extinction, thereby converting all input material of the refining system **100** into aviation fuel or lighter products. Additionally, the fractionator **120** includes dual functions of separating the initial renewable diesel feedstock **118** into fractions, as well as separating the hydrocracked product **132** into the fractions.
(27) From the fractionator **120**, the aviation fuel-range fraction **124** of C.sub.8-18 compounds is supplied to the isomerization reactor **140** to be further processed or refined into an aviation fuel product **142**. In some examples, the aviation fuel product **142** is an SPK product that is blended with an aromatic-containing fuel, such as an aromatic sustainable fuel and/or a petroleum fuel, to form an SAF having various standardized properties. The aromatic sustainable fuel of certain examples is obtained from processing of a sustainable source (e.g., sugars) and/or from an aromatic chemical production facility. As an example, a predominantly paraffinic SPK product may be blended with the aromatic-containing fuel to produce SAF containing a minimum content of aromatic compounds, as specified by one or more ASTM standards. As another example, to meet certain ASTM standards for aviation turbine fuel (D1655 and D7566), an aviation fuel may constitute up to 50% of the SPK product that is blended with a petroleum-derived jet fuel to produce the aviation fuel with final physical and chemical properties that correspond to standardized aviation use. In certain examples, the aviation fuel product **142** is used directly as an SAF without blending with an aromatic-containing fuel. The isomerization reactor **140** includes a third vessel that has a third inlet and a fifth outlet. In examples, the third inlet of the isomerization reactor **140** is connected to and in fluid communication with the second outlet of the fractionator **120** to receive at least a portion of the aviation fuel-range fraction **124**. The isomerization reactor **140** receives an entirety of the aviation fuel-range fraction **124** or a portion thereof, in certain examples. The aviation fuel-range fraction **124** of certain examples also includes C.sub.8-18 compounds of the hydrocracked product **132** provided to the fractionator **120** from the hydrocracking reactor **130**, which enhances product yield.
(28) The isomerization reactor **140** isomerizes the C.sub.8-18 compounds of the aviation fuel-range fraction **124** into the aviation fuel product **142**, which is output to the fifth outlet of the isomerization reactor **140**. For example, operating conditions in the isomerization reactor **140** are selected to lower a freeze point of the aviation fuel-range fraction **124** to meet aviation fuel specifications by converting normal alkanes into branched or iso-alkanes of the same molecular weight. The isomerization reactor **140** isomerizes compounds with an isomerization catalyst in the presence of hydrogen, which is provided by any suitable hydrogen source. For example, the partial pressure of hydrogen in the isomerization reactor **140** is at least 200 psi, in some examples. In some examples, the partial pressure of hydrogen is at least 300, 400, 500, or more psi. The isomerization reactor **140** and the hydrocracking reactor **130** may receive hydrogen from separate hydrogen sources or from a shared hydrogen source, in examples. In some examples, the isomerization reactor **140** includes an operating temperature in a range between about 400 and about 800° F.

(between about 204.4 and about 426.7° C.). In some examples, the operating temperature is in a range between about 300 and about 700° F. (between about 148.9 and about 371.1° C.). In some examples, the operating temperature is in a range between about 500 and about 900° F. (between about 260.0 and about 482.2° C.).

(29) The particular configuration of the refining system **100** further enhances the operation of the isomerization reactor **140**. For example, because only C.sub.8-18 compounds are isomerized, lower severity operation and a smaller catalyst volume are be used in the isomerization reactor **140**, compared to systems that may also isomerize light ends or C.sub.8− compounds. The refining system **100** may also exclude or avoid a stabilizer for adjusting an initial boiling point of the aviation fuel product **142**, based on the specific separation of the aviation fuel-range fraction **124** from other components in the fractionator **120**. The reduced demand for a stabilizer further enables the isomerization reactor **140** to be constructed with a smaller size and corresponding reduced capital cost. Removal of the C.sub.8− compounds prior to processing also facilitates maintenance of a higher hydrogen partial pressure within the isomerization, thus providing both reduced deactivation of an isomerization catalyst and improved isomerization selectivity for the aviation fuel product **142**.

(30) FIG. **2** is an illustration of a method **200** for producing aviation fuel, according to some embodiments disclosed herein. The method **200** is described with reference to the refining system **100** of FIG. **1**. Additionally, the steps or actions of the method **200** may be completed, implemented, or controlled by a suitable control component, such as a controller **502** discussed with reference to FIG. **5**. For example, the method **200** may be included in one or more programs, protocols, or instructions loaded into a memory **506** of the controller **502** and executed on one or more processors **504** of the controller **502**.

(31) At block **202** of the method **200**, the refining system **100** (or a controller thereof) supplies a renewable diesel feedstock **118** to a fractionator **120**. The renewable diesel feedstock **118** is a HDO or hydrodeoxygenated renewable diesel feedstock that contains normal alkanes produced from triglycerides, in certain examples. At block **204**, the refining system **100** fractionates the renewable diesel feedstock into a light fraction **122**, an aviation fuel-range fraction **124**, and a heavy fraction **126**. As noted above, the fractionator **120** may include a first inlet to receive the renewable diesel feedstock, a first outlet to output the light fraction **122**, a second outlet to output the aviation fuel-range fraction **124**, and a third outlet to output the heavy fraction **126**.

(32) At block **206**, the refining system **100** supplies the heavy fraction **126** to a hydrocracking reactor **130**. Thus, at block **208**, the refining system **100** hydrocracks the heavy fraction **126** into a hydrocracked product **132**. At block **210**, the refining system **100** recycles the hydrocracked product **132** to the fractionator **120**. The hydrocracking reactor **130** may include a second inlet to receive the heavy fraction **126** and a fourth outlet to supply the hydrocracked product **132** to the fractionator **120**. As such, a recycle loop is provided to circulate the heavy fraction **126** through the hydrocracking reactor **130** and the fractionator **120** to produce smaller compounds that further increase the yield of aviation fuel.

(33) Moreover, at block **212**, the refining system **100** supplies the aviation fuel-range fraction **124** to an isomerization reactor **140**. In certain examples, block **206** and block **212** are performed in parallel, such as during operations in which the refining system is operated in a continuous processing mode. At block **214**, the refining system **100** isomerizes the aviation fuel-range fraction **124** into an aviation fuel product **142**. For example, the isomerization reactor **140** may include a third inlet to receive the aviation fuel-range fraction **124** and a fifth outlet to output the aviation fuel product **142**. As such, at block **216**, the refining system **100** outputs the aviation fuel product **142** from the fifth outlet to be further utilized in or transported from the refining system **100**.

(34) FIG. **3** is a schematic diagram of another refining system **300** for producing aviation fuel, according to some embodiments disclosed herein. As illustrated, the refining system **300**, for example, includes two recycle loops that cooperate to improve the yield, efficiency, and/or

properties of aviation fuel produced by the refining system **300**. The refining system **300** of FIG. **3** includes certain components or units that are similar to those discussed above with reference to FIG. **1**. These components or units are similarly labeled, and their descriptions are not repeated in detail for improved clarity. For example, the refining system **300** includes a fractionator **320**, a hydrocracking reactor **330**, and an isomerization reactor **340**.
(35) As discussed above, the fractionator **320** may include a first vessel that has a first inlet, a first outlet, a second outlet, and a third outlet. The fractionator **320** separates the renewable diesel feedstock **318** into three fractions: a light fraction **322** having substantially C.sub.1-8 or C.sub.8− compounds, an aviation fuel-range fraction **324** having substantially C.sub.8-18 compounds, and a heavy fraction **326** having substantially C.sub.18+ compounds. The hydrocracking reactor **330** may include a second vessel that has a second inlet to receive at least a portion of the heavy fraction **326** and a fourth outlet. Additionally, the isomerization reactor **340** may include a third vessel that has a third inlet to receive at least a portion of the aviation fuel-range fraction **324** and a fifth outlet. To illustrate another non-limiting example, the refining system **300** of FIG. **3** includes a fourth inlet into the fractionator **320** connected to and in fluid communication with the fourth outlet of the hydrocracking reactor **330** to receive at least a portion of a hydrocracked product **332** therefrom. The fourth inlet may be independent of the first inlet associated with the renewable diesel feedstock **318**, thus providing an additional control variable for managing operation of the fractionator **320**. In some examples, the fourth inlet is provided at a different tray or height of the fractionator **320**, compared to the first inlet.
(36) As an additional feature, the refining system **300** includes a separator **350** positioned downstream of the isomerization reactor **340** to further refine the aviation fuel produced by the refining system **300**. For example, certain operations or examples of the isomerization reactor **340** generate additional C.sub.8− light ends or compounds via cracking during isomerization of the aviation fuel-range fraction **324** having C.sub.8-18 compounds. The fifth outlet of the isomerization reactor **340** therefore outputs a mixed product **348** having C.sub.17− compounds. The separator **350** includes a fourth vessel having a fifth inlet, a sixth outlet, and a seventh outlet. The fifth inlet is connected to and in fluid communication with the fifth outlet of the isomerization reactor **340** to supply at least a portion of the mixed product **348** to the fourth vessel. In certain examples, the separator **350** is a flash drum or flash that performs separations based on flash evaporation of the mixed product **348**. In some examples, the separator **350** includes a stripper, as well as an additional inlet that receives a suitable stripping gas for facilitating the desired separation therein.
(37) Embodiments of the separator **350** include suitable operating conditions to separate or strip a produced light fraction **352** of C.sub.8− compounds, or additional C.sub.8− fraction, from the aviation fuel product of C.sub.8-18 compounds. Accordingly, the sixth outlet at a top of the separator **350** is connected to recycle at least a portion of the produced light fraction **352** back to the fractionator **320**, from which the produced light fraction **352** are directed through the first outlet of the fractionator **320** as the light fraction **322**. Certain examples of the fractionator **320** include a sixth inlet that is connected to and in fluid communication with the sixth outlet of the separator **350**. In other examples, the produced light fraction **352** may be mixed with the renewable diesel feedstock **318**. Additionally, the seventh outlet at a bottom of the separator **350** outputs the relatively heavier aviation fuel product **342**, which includes C.sub.8-18 compounds within targeted aviation fuel specifications.
(38) FIG. **4** is an illustration of a method **400** for producing aviation fuel, according to some embodiments disclosed herein. The method **400**, for example, is described with reference to the refining system **300** of FIG. **3**. Additionally, the steps or actions of the method **400** may be completed, implemented, or controlled by a suitable control component, such as a controller **502** discussed with reference to FIG. **5**. For example, the method **400** may be included in one or more programs, protocols, or instructions loaded into a memory **506** of the controller **502** and executed

on one or more processors **504** of the controller **502**.
(39) The method **400** includes certain blocks or actions that are similar to those of method **200** discussed above with reference to FIG. **2**. These blocks, including blocks **402**, **404**, **406**, **408**, **410**, and **412**, and their descriptions are not repeated here for improved clarity. Looking to block **420**, the refining system **300** isomerizes the aviation fuel-range fraction **324** into an aviation fuel product **342** and a produced light fraction **352**. For example, as discussed above, certain operations of the isomerization reactor **340** cause production of additional C.sub.8− compounds. At block **422**, the refining system **300** separates the aviation fuel product **342** from the produced light fraction **352**. Certain examples include performing the separation by stripping the produced light fraction **352** from the aviation fuel product **342** within a separator **350**.
(40) At block **424**, the refining system **300** recycles the produced light fraction **352** to the fractionator **320**. As such, a second recycle loop is provided within the refining system to enable further refinement of the aviation fuel product **342** at desirably increased yields. At block **426**, the refining system **300** outputs the aviation fuel product **342** to be used in and/or transported from the refining system **300**.
(41) FIG. **5** is a simplified diagram illustrating a control system **500** for managing the producing of aviation fuel according to some embodiments disclosed herein. In some examples, the control system **500** includes a controller **502** or one or more controllers. Certain examples include the controller **502** being in signal communication with various other controllers throughout or external to a refining system, such as one of the refining systems **100**, **300** discussed above. Additionally, the controller **502** may be considered a supervisory controller or other suitable control component for managing the refining system, as discussed herein.
(42) The controller **502** of various examples disclosed herein include one or more processors, such as processor **504**, as well as a memory or machine-readable storage medium, such as memory **506**. As used herein, a “machine-readable storage medium” may be any electronic, magnetic, optical, or other physical storage apparatus to contain or store information such as executable instructions, data, and the like. For example, any machine-readable storage medium described herein may be any of random access memory (RAM), volatile memory, non-volatile memory, flash memory, a storage drive, a hard drive, a solid state drive, any type of storage disc, and the like, or a combination thereof. The memory **506** stores or includes instructions executable by the processor **504**. As used herein, a “processor” includes, for example, one processor or multiple processors included in a single device or distributed across multiple computing devices. The processor **304** may be at least one of a central processing unit (CPU), a semiconductor-based microprocessor, a graphics processing unit (GPU), a field-programmable gate array (FPGA) to retrieve and execute instructions, a real time processor (RTP), other electronic circuitry suitable for the retrieval and execution instructions stored on a machine-readable storage medium, or a combination thereof.
(43) As used herein, “signal communication” refers to electric communication such as hard wiring two components together or wireless communication, as understood by those skilled in the art. For example, wireless communication may be Wi-Fi®, Bluetooth®, ZigBee, or forms of near field communications. In addition, signal communication may include one or more intermediate controllers or relays disposed between elements that are in signal communication with one another. In the drawings and specification, several examples of systems and methods of producing aviation fuel are disclosed. The controller **502** includes instructions **508** to produce the aviation fuel according to the examples disclosed herein.
(44) In some examples, the instructions **508** cause the refining system to produce the aviation fuel product. For example, the controller **502** may receive any suitable product specifications and/or operating conditions from a user interface **510** and/or from data pre-loaded into the memory **506**. To produce the aviation fuel product to the desired or target specifications, the controller **502** sends signals to and/or receives signals from a fractionator **520**, a hydrocracking reactor **530**, an isomerization reactor **540**, and/or any other units associated with the refining system. The signals

transmitted by the controller **502** may include operating parameters for each unit, such as temperatures, length of operating time, amount of incoming feedstock, amount of product to be produced, as well as other parameters. As an example, the controller **502** selects an operating temperature for the hydrocracking reactor **530**, which is lower than a reference operating temperature of a reference hydrocracking reactor that is downstream instead of upstream of an associated reference isomerization reactor. Additionally, the controller **502** selects an operating temperature for the isomerization reactor **540** that is lower than a reference operating temperature of a reference isomerization reactor that is upstream instead of downstream of an associated reference hydrocracking reactor. The instructions **508** and/or any other suitable portion of the controller **502** may include any suitable software applications, routines, or programming to produce the aviation fuel product. For examples, the instructions **508** can include a fractionating module **521** for controlling operation of the fractionator **520**, a hydrocracking module **531** for controlling operation of the hydrocracking reactor **530**, and an isomerizing module **541** for controlling operation of the isomerization reactor **540**.

(45) The control system **500** of some examples may also include various sensors and meters disposed through the refining system, such as the illustrated sensors **550** communicatively coupled to the controller **502**. The sensors and meters may be in signal communication with the controller **502** and may provide data or feedback to the controller **502** to determine various properties of each unit and/or product at various stages in the process. The sensors and meters may measure flow, density, chemical properties, temperature, pressure, and/or other properties, utilized for monitoring the production of the aviation fuel product. Based on data received from the sensors **550**, the controller **502** can monitor the process from initiation through completion to produce the aviation fuel product. In some examples, the control system **500** also includes various actuators **560** in signal communication with the controller **502** to facilitate adjustment and/or control of the various operating parameters, properties, and so forth. The actuators **560** can include any suitable pneumatic, electric, mechanical, and/or hydraulic control components, including control valves, pumps, compressors, heating elements, electrical switches, and so forth. The controller **502** can instruct one or more actuators **560** on demand and/or at a predetermined or selected schedule to adjust flows, densities, chemical properties, temperatures, pressures, and/or other properties utilized during production of the aviation fuel product.

(46) FIG. **6** is an illustration of a method **600** for controlling production of aviation fuel, according to some embodiments disclosed herein. The method **600**, for example, is described with reference to the control system **500** of FIG. **5**. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the method. Additionally, the steps or actions of the method **600** may be completed, implemented, or controlled by the controller **502** discussed above. The method **600** may be included in one or more modules, programs, protocols, or instructions of the controller **502**. For example, certain steps can be performed by one or more of the fractionating modules **521**, the hydrocracking module **531**, and the isomerizing module **541**, respectively. As a non-limiting example, the fractionating module **521** may perform the steps of the left-hand column of the method **600**, the hydrocracking module **531** may perform the steps of the center column of the method **600**, and the isomerizing module **541** may perform the steps of the right-hand column of the method **600**. In certain examples, each column of the method **600** is performed in parallel, such as via the various modules of the controller **502**, which can communicate with one another as described below.

(47) As recognized herein, the fractionator **520**, the hydrocracking reactor **530**, and the isomerization reactor **540** work cooperatively to produce aviation fuel in a system arranged as discussed above, in which operation of an upstream component can influence operation of any downstream component. The present example of the method **600** illustrates certain control steps that can improve the interoperation of equipment arranged according to the present disclosure, such

as by allowing specialization of control components and/or more rapid detection and/or prevention of any undesired operations or products by the fractionator **520**, the hydrocracking reactor **530**, and the isomerization reactor **540**.

(48) Looking first to the left-hand column, at block **620** of the method **600**, the controller **502** receives sensor data indicative of operation of the fractionator **520**. For example, the controller **502** may receive one or more signals indicative of flow, density, chemical properties, temperature, pressure, and/or other properties associated with operation of the fractionator **520**. At block **622**, the controller **502** determines whether the fractionator **520** is operating within one or more preselected operating parameter ranges. In some examples, the operating parameter ranges may be preselected and/or updated on demand in response to instructions received from a user interface or device, from another controller, and/or from another module of the control system. The controller **502** can evaluate each received operating parameter individually or collectively to identify any operating parameters that are currently deviating, or predicted to deviate in the near future, from their corresponding ranges. In response to determining that the fractionator **520** is operating within the preselected operating parameter ranges or that no deviations are present, the controller **502** returns to block **620** to continue monitoring the sensor data associated with the fractionator **520**.

(49) In response to determining that, at block **622**, the fractionator **520** includes one or more operating parameters that are not within their associated preselected ranges, the controller **502** proceeds to block **626** to determine whether a deviation is based on operation of the hydrocracking reactor **530**. That is, the fractionator **520** receives the renewable diesel feedstock and a recycle stream of hydrocracked product from the hydrocracking reactor **530**. Based on the sensor data, the controller **502** can determine whether the deviation in the fractionator **520** originates with or is based at least in part on the hydrocracked product from the hydrocracking reactor **530**. In response to determining that, at block **626**, the deviation is not based on operation of the hydrocracking reactor **530**, the controller **502** continues to block **628** to instruct one or more actuators to adjust operation of the fractionator **520**. The controller **502** can thus correct the fractionator **520** to reduce or eliminate the deviation, before continued monitoring of the fractionator **520** at block **620**.

(50) In response to determining that, at block **626**, the deviation is based on operation of the hydrocracking reactor **530**, the controller **502** proceeds to block **638** to instruct one or more actuators to adjust operation of the hydrocracking reactor **530**. In some examples in which the fractionating module **521** performs steps of the left-hand column of the method **600**, the fractionating module **521** provides one or more signals to the hydrocracking module **531** (as illustrated by a dash-dotted line) to enable the hydrocracking module **531** to perform the adjustments to the hydrocracking reactor **530**.

(51) Turning to the center column of the method **600**, at block **630**, the controller **502** receives sensor data indicative of operation of the hydrocracking reactor **530**. As discussed in detail above, the hydrocracking reactor **530** is in a recycle loop with the fractionator **520**, which can each affect one another during operation. Additionally, the hydrocracking reactor **530** and the fractionator **520** are both upstream of the isomerization reactor **540**, which does not affect operation of the hydrocracking reactor **530** and the fractionator **520**. At block **632**, the controller **502** determines whether the hydrocracking reactor **530** is operating within one or more preselected operating parameter ranges. As noted above, the controller **502** can monitor any suitable operating parameter associated with operation of the hydrocracking reactor **530**, and the operating parameter ranges may be preselected and/or updated on demand in response to instructions received from any suitable device or module. In response to determining that the hydrocracking reactor **530** is operating within the preselected operating parameter ranges or that no deviations are present, the controller **502** returns to block **630** to continue monitoring the sensor data associated with the hydrocracking reactor **530**.

(52) In response to determining that the hydrocracking reactor **530** includes one or more operating parameters that are not within their associated preselected ranges, the controller **502** proceeds to

block **634** to determine whether a deviation is based on operation of the fractionator **520**. In response to determining that, at block **634**, the deviation is based on operation of the fractionator **520**, the controller **502** proceeds to block **628** to instruct one or more actuators to adjust operation of the fractionator **520**. In some examples in which the hydrocracking module **531** performs steps of the center column of the method **600**, the hydrocracking module **531** provides one or more signals to the fractionating module **521** (as illustrated by a dashed line) to enable the fractionating module **521** to perform the adjustments to the fractionator **520**.

(53) In response to determining that, at block **634**, the deviation is not based on operation of the fractionator **520**, the controller **502** continues to block **638** to instruct one or more actuators to adjust operation of the hydrocracking reactor **530**. The controller **502** can thus correct the hydrocracking reactor **530** to reduce or eliminate the deviation, before continued monitoring of the hydrocracking reactor **530** at block **630**.

(54) With focus on the right-hand column of the method **600**, at block **640**, the controller **502** receives sensor data indicative of operation of the isomerization reactor **540**. As discussed in detail above, the isomerization reactor **540** is downstream of both the fractionator **520** and the hydrocracking reactor **530**, which each affect operation of the isomerization reactor **540**. At block **642**, the controller **502** determines whether the isomerization reactor **540** is operating within one or more preselected operating parameter ranges. As noted above, the controller **502** can monitor any suitable operating parameter associated with operation of the isomerization reactor **540**, and the operating parameter ranges may be preselected and/or updated on demand in response to instructions received from any suitable device or module. In response to determining that the isomerization reactor **540** is operating within the preselected operating parameter ranges or that no deviations are present, the controller **502** returns to block **640** to continue monitoring the sensor data associated with the isomerization reactor **540**.

(55) In response to determining that the isomerization reactor **540** includes one or more operating parameters that are not within their associated preselected ranges, the controller **502** proceeds to block **644** to determine whether a deviation is based on operation of the fractionator **520**. In response to determining that the deviation is based on operation of the fractionator **520**, the controller **502** proceeds to block **628** to instruct one or more actuators to adjust operation of the fractionator **520**. In some examples in which the isomerizing module **541** performs steps of the right-hand column of the method **600**, the isomerizing module **541** provides one or more signals to the fractionating module **521** (as illustrated by a dashed line) to enable the fractionating module **521** to perform the adjustments to the fractionator **520**.

(56) In response to determining that, at block **644**, the deviation is not based on operation of the fractionator **520**, the controller **502** continues to block **646** to determine whether the deviation is based on operation of the hydrocracking reactor **530**. In response to determining that the deviation is based on operation of the hydrocracking reactor **530**, the controller **502** proceeds to block **638** to instruct one or more actuators to adjust operation of the hydrocracking reactor **530**. For examples in which the isomerizing module **541** performs steps of the right-hand column of the method **600**, the isomerizing module **541** provides one or more signals to the hydrocracking module **531** (as illustrated by a dash-dotted line) to enable the hydrocracking module **531** to perform the adjustments to the hydrocracking reactor **530**.

(57) In response to determining that, at block **646**, the deviation is not based on operation of the fractionator **520** or the hydrocracking reactor **530**, the controller **502** continues to block **648** to instruct one or more actuators to adjust operation of the isomerization reactor **540**. The controller **502** can thus correct the isomerization reactor **540** to reduce or eliminate the deviation, before continued monitoring of the isomerization reactor **540** at block **640**. Accordingly, the controller **502** performing the method **600** enables each of the fractionator **520**, the hydrocracking reactor **530**, and the isomerization reactor **540** to operate in an improved and cohesive manner, leveraging their arrangement to facilitate production of aviation fuel at higher efficiencies and yields than

previously available systems.
(58) This application claims priority to, and the benefit of U.S. Provisional Application No. 63/548,081, filed Nov. 10, 2023, titled "SYSTEMS AND METHODS FOR PRODUCING AVIATION FUEL," the disclosure of which is incorporated herein by reference in its entirety.
(59) Although specific terms are employed herein, the terms are used in a descriptive sense only and not for purposes of limitation. Embodiments of systems, methods, and controllers have been described in considerable detail with specific reference to the illustrated examples. However, it will be apparent that various modifications and changes can be made within the spirit and scope of the embodiments of systems, methods, and controllers as described in the foregoing specification, and such modifications and changes are to be considered equivalents and part of this disclosure.

Claims

1. A method to produce aviation fuel, the method comprising: fractionating a renewable diesel feedstock in a fractionator to produce a C.sub.8− fraction, a C.sub.8-18 fraction, and a C.sub.18+ fraction; providing the C.sub.8-18 fraction to an isomerization reactor to produce an aviation fuel product; supplying at least a portion of the C.sub.18+ fraction to a hydrocracking reactor to produce a hydrocracked product; and recycling at least a portion of the hydrocracked product to the fractionator for fractionating along with the renewable diesel feedstock.
2. The method of claim 1, wherein the renewable diesel feedstock comprises a hydrodeoxygenated renewable diesel.
3. The method of claim 1, wherein the renewable diesel feedstock contains normal alkanes produced by hydrodeoxygenation of glycerides.
4. The method of claim 1, further comprising hydrotreating a renewable feedstock containing triglycerides, triglyceride derivatives, or a combination thereof in a hydrogen-rich atmosphere to produce the renewable diesel feedstock.
5. The method of claim 1, further comprising: mixing the hydrocracked product with the renewable diesel feedstock directly upstream of the fractionator.
6. The method of claim 1, wherein the renewable diesel feedstock is supplied to a first inlet of the fractionator, and wherein the at least a portion of the hydrocracked product is supplied to a second inlet of the fractionator.
7. The method of claim 1, wherein the recycling at least a portion of the hydrocracked product to the fractionator further comprises recycling an entirety of the hydrocracked product.
8. A method to produce aviation fuel, the method comprising: fractionating a renewable diesel feedstock in a fractionator to produce a C.sub.8− fraction, a C.sub.8-18 fraction, and a C.sub.18+ fraction; providing the C.sub.8-18 fraction to an isomerization reactor to produce an aviation fuel product; supplying at least a portion of the C.sub.18+ fraction to a hydrocracking reactor to produce a hydrocracked product; recycling at least a portion of the hydrocracked product to the fractionator for fractionating along with the renewable diesel feedstock; producing an additional C.sub.8− fraction in the isomerization reactor; providing the aviation fuel product and the additional C.sub.8− fraction to a separator to separate the aviation fuel product from the additional C.sub.8− fraction; and providing the additional C.sub.8− fraction to the fractionator.
9. The method of claim 1, further comprising: selecting an operating temperature of the hydrocracking reactor, the operating temperature being lower than a reference operating temperature of a reference hydrocracking reactor downstream of a reference isomerization reactor.
10. A method to produce aviation fuel, the method comprising: supplying a hydrodeoxygenated renewable diesel feedstock to a fractionator; fractionating the hydrodeoxygenated renewable diesel feedstock into a C.sub.8− fraction, a C.sub.8-18 fraction, and a C.sub.18+ fraction; providing the C.sub.8-18 fraction to an isomerization reactor to produce an aviation fuel product; supplying at least a portion of the C.sub.18+ fraction to a hydrocracking reactor to produce a hydrocracked

product, the hydrocracking reactor cracking one or more heavy compounds of the C.sub.18+ fraction into two or more lighter compounds; and recycling at least a portion of the hydrocracked product to the fractionator for fractionating along with the hydrodeoxygenated renewable diesel feedstock.

11. The method of claim 10, further comprising: hydrodeoxygenating a triglyceride feedstock to produce the hydrodeoxygenated renewable diesel feedstock.

12. The method of claim 10, further comprising: producing an additional C.sub.8− fraction in the isomerization reactor; providing the aviation fuel product and the additional C.sub.8− fraction to a separator to separate the aviation fuel product from the additional C.sub.8− fraction; and providing the additional C.sub.8− fraction to the fractionator.

13. A system to produce aviation fuel, the system comprising: a fractionator having a first inlet to receive a renewable diesel feedstock, a first outlet, a second outlet, and a third outlet, the fractionator operable to fractionate the renewable diesel feedstock into a C.sub.8− fraction output through the first outlet, a C.sub.8-18 fraction output to the second outlet, and a C.sub.18+ fraction output to the third outlet; a hydrocracking reactor having a second inlet and a fourth outlet, the second inlet connected to and in fluid communication with the third outlet to receive the C.sub.18+ fraction, the hydrocracking reactor operable to hydrocrack the C.sub.18+ fraction into a hydrocracked product, and the fourth outlet connected to recycle the hydrocracked product to the fractionator; and an isomerization reactor having a third inlet and a fifth outlet, the third inlet connected to and in fluid communication with the second outlet to receive the C.sub.8-18 fraction, and the isomerization reactor operable to isomerize the C.sub.8-18 fraction into an aviation fuel product output to the fifth outlet.

14. The system of claim 13, wherein the renewable diesel feedstock contains normal alkanes produced by hydrodeoxygenation of glycerides, free fatty acids, or combinations thereof.

15. The system of claim 13, wherein a majority of the hydrodeoxygenated renewable diesel contains C.sub.16 and C.sub.18 normal alkanes.

16. The system of claim 13, further comprising a separator having a fifth inlet connected to and in fluid communication with the fifth outlet, a sixth outlet to supply an additional C.sub.8− fraction to the fractionator, and a seventh outlet to output the aviation fuel product.

17. The system of claim 13, wherein the fourth outlet is connected to and in fluid communication with the first inlet or an additional inlet of the fractionator.

18. The system of claim 13, further comprising a hydrogen source connected to and in fluid communication with the hydrocracking reactor, the isomerization reactor, or both.

19. The system of claim 13, further comprising a controller in signal communication with the fractionator, the hydrocracking reactor, and the isomerization reactor, and wherein the controller is configured to transmit signals to one or more of the fractionator, the hydrocracking reactor, or the isomerization reactor to adjust production of the aviation fuel product.

20. The system of claim 19, wherein the controller is configured to adjust operation of the fractionator, the hydrocracking reactor, or both in response to detecting that the isomerization reactor is deviating from one or more preselected operating parameter ranges.